

Conditional diffusion emulator of Veros ACC equilibrium states: first end-to-end results

21 September 2026

Summary

- Goal: generate the post-spin-up (T, S) state of the Veros ACC channel directly from the EKE-closure coefficients (c_k, c_ϵ) , skipping spin-up.
- Pipeline: DINO-Fusion diffusion code adapted to a parameter ensemble; runs end to end on Jean Zay in under one A100 hour.
- Result: with per-level normalisation the hold-out error is 0.19–0.21 K on every split, at or above the mean-state baseline in the interior and above the nearest training run on average; it beats copying the row below only on the warm side of the extrapolated row. A Wasserstein metric on the stratification profiles gives the same ranking and shows that training and held-out conditions score alike.
- The sampler’s clip matters as much as the split: a scalar clip at $|x'| \leq 1$ gives 0.15 / 0.14 / 0.17 K but caps the warm corner; the per-level clip used here frees the corner and costs 0.03–0.06 K, all of it at the bottom levels (Table 1).
- Physical miss: the model generates temperature inversions at the rate of the weakly eddying runs in every regime.

1 Data

- 100 Veros ACC runs, 10×10 log-spaced grid: $c_k \in [0.0125, 0.8]$, $c_\epsilon \in [0.0875, 5.6]$; 100 years, 30-day output, interior grid $30 \times 42 \times 15$ (x, y, z).
- Kept: T and S , last 20 years (241 snapshots/run). $S = 35$ psu everywhere in every run.
- Nearly deterministic end states: within-window std 0.01 K above 1400 m vs 0.05 K between grid neighbours; bottom level still drifts by 0.45 K over the window.
- One effective parameter: $\log c_k - 1.05 \log c_\epsilon$ explains 99 % of the variance of the domain-mean T ; the low- c_k/c_ϵ part of the grid saturates.

2 Method

- Fields $\rightarrow$ 30 channels $(T_{1..15}, S_{1..15})$, normalised per level, land $\rightarrow 0$, zero-padded $42 \times 30 \rightarrow 48 \times 32$:

$$x'_z = \frac{x_z - \mu_z}{3 \max(\sigma_z, 0.05)}. \quad (1)$$

μ_z, σ_z are computed once on all 100 runs and held fixed for every hold-out split (a deliberate, mild leakage of 30 scaling constants; no held-out state enters training). At every sampling step the predicted clean state is clipped per channel to the observed data range of that level over all runs, widened by 5 % (bounds within $[-1.30, 2.86]$; a scalar clip at $|x'| \leq 1$ would cut the deep levels, whose real values reach 2.7).

- Denoiser: UNet2DModel (64, 64, 128, 128), 6.9 M parameters. The condition c , the standardised pair $(\log c_k, \log c_\epsilon)$, enters through an MLP added to the timestep embedding.

- DDPM, ϵ -prediction, 1000 steps, squared-cosine schedule, EMA:

$$\mathcal{L}(\theta) = \mathbb{E}_{t, x_0, \epsilon} \left\| \epsilon - \epsilon_\theta(\sqrt{\bar{\alpha}_t} x_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon, t, c) \right\|^2. \quad (2)$$

20 000 steps, batch 32, AdamW 10^{-4} with cosine decay; 22 min on one A100.

- Sampling: 1000-step DDPM loop with EMA weights; land and padding re-zeroed after every step; 32 samples per condition.

3 Evaluation

- Truth: 20-year time-mean state of each held-out run; metrics on water cells.
- Baselines: mean training state; nearest training run in standardised log-parameter space; average of the axis-neighbour training runs on the grid.
- Three hold-out splits, same data file and statistics, different training sets: scattered (10 interior points), band of three rows ($c_k = 0.126, 0.2, 0.3175$; 30 runs; training rows a factor 6.3 apart), top row ($c_k = 0.8$; 10 runs; extrapolation).
- Distributional metric. Each state is reduced to its profile of horizontal means over water cells, $\bar{T}_z$. Per level, the Wasserstein-1 distance between the 32 generated values and all 241 true snapshots of the window, through the quantile functions,

$$W_1^{(z)} = \int_0^1 |F_{g,z}^{-1}(q) - F_{t,z}^{-1}(q)| dq, \quad W_1 = \sum_z \frac{\Delta z_z}{H} W_1^{(z)}, \quad (3)$$

with Δz_z the level thicknesses from the level midpoints ($H = 2080$ m). All 241 snapshots are used, not a subsample of 32: the quantile form accepts unequal sizes and a subsample would only add noise. A point prediction x (the baselines) gives $W_1^{(z)} = \frac{1}{241} \sum_i |x_z - \bar{T}_{z,i}|$. Unlike the RMSE of the ensemble mean, this rewards a correct spread and does not reward collapse to the mean; it ignores horizontal structure, which the RMSE keeps.

	scattered 10 runs	band of three rows 30 runs	top row 10 runs
diffusion, ensemble mean of 32: RMSE (K)	0.197	0.194	0.206
diffusion, single sample: RMSE (K)	0.213	0.212	0.227
nearest training run: RMSE (K)	0.037	0.116	0.170
neighbour average: RMSE (K)	0.020	0.116	0.170
mean training state: RMSE (K)	0.155	0.197	0.470
diffusion: domain-mean bias (K)	-0.011	-0.010	-0.032
ensemble spread / true spread (K)	0.088 / 0.031	0.088 / 0.031	0.099 / 0.032
T inversions (interfaces), generated / truth	8.9 % / 7.7 %	10.3 % / 1.3 %	12.2 % / 0.1 %
max salinity error (psu)	0.003	0.003	0.003
W_1 profile, diffusion (K)	0.161	0.156	0.158
W_1 profile, neighbour average (K)	0.036	0.095	0.142
W_1 profile, nearest training run (K)	0.044	0.095	0.142
W_1 profile, mean training state (K)	0.136	0.160	0.420
same models, scalar clip $ x' \leq 1$: RMSE (K)	0.152	0.135	0.173
same models, scalar clip $ x' \leq 1$: W_1 (K)	0.127	0.111	0.143

Table 1: Hold-out metrics per split, mean over the held-out runs, water cells only, 32 samples per run, per-level clip. Last two rows: the same trained models sampled with the scalar clip.

4 Findings

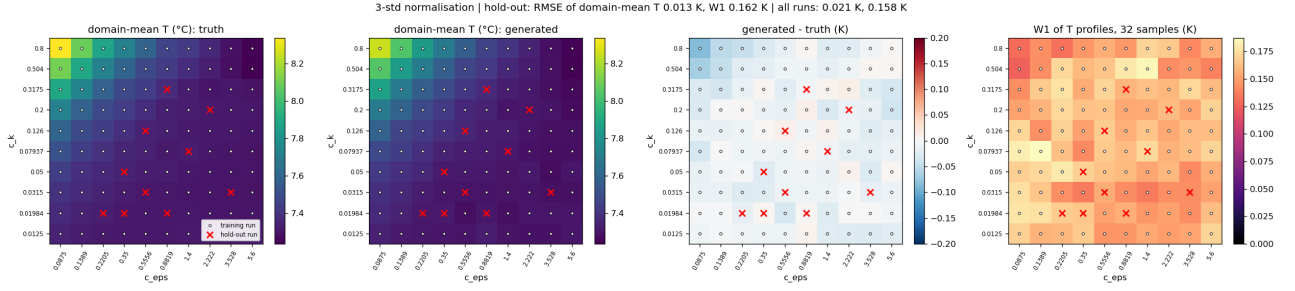
- **Conditioning is weak under per-level scaling.** Signal-to-range ratio $\rho_z = \sigma_z^{\text{runs}} / (3\sigma_z)$ is 0.7 % at the surface, 9 % at 650 m, 25 % at the bottom; the ϵ -loss weights all cells equally, so the shared structure dominates. Scattered split: RMSE $\geq$ mean-state baseline (Figure 1).

- **Model error is split-independent (0.19–0.21 K); baselines degrade with the gap (0.02–0.17 K).** The model loses to the nearest run in all three band rows (0.20 / 0.20 / 0.18 K vs 0.06 / 0.13 / 0.15 K for $c_k = 0.126 / 0.2 / 0.3175$); on the top row it beats copying the row below on the warm side ($c_\epsilon \leq 0.35$: 0.19–0.29 vs 0.22–0.35 K, corner included), ties at $c_\epsilon = 0.56$ and loses in the flat regime (Figures 2, 3).
- **Depth structure.** Beats the nearest run between 370 and 800 m in the band split (0.10 K vs 0.14 K at 650 m) and between 260 and 1200 m in the top row; worse than baselines in the top 250 m (scale $3\sigma_z \approx 12$ K amplifies sampler noise) and at the two bottom levels (skewed, drifting training distribution, and the widest clip bounds).
- **The sampler’s clip is a regulariser.** Same trained models, sampling only, scattered / top row: scalar clip at $|x'| \leq 1$ gives 0.152 / 0.173 K, at 1.5 it gives 0.194 / 0.183 K, at 3 it gives 0.302 / 0.277 K, although the final samples barely exceed $|x'| = 1$ (0.1 % of values): clipping the early, inaccurate clean-state estimates keeps the chain on track. A scalar clip at 1 also caps the deep levels of the warm corner (12 % of the top row’s true bottom values), a floor of 0.04 K on the row and 0.17 K at the corner run. The per-level clip used here removes that floor (corner run 0.29 vs 0.40 K, top-row bias -0.03 vs -0.07 K) and cuts the salinity error by four, but loosens the regulariser where the bounds are wide: bottom-level RMSE $0.34 \rightarrow 0.48$ K (scattered), hence 0.197 / 0.194 / 0.206 K overall, worse than the scalar clip on every split (Table 1). Above 1400 m the two clips are within 0.04 K of each other.
- **Inversions.** Generated 9–12 % of interfaces with $\partial_z T < 0$ in every regime; truth: 7.7 % (scattered) $\rightarrow$ 1.3 % (band) $\rightarrow$ 0.1 % (top). The regime dependence is not learned.
- **W_1 on the mean profiles agrees with the RMSE ranking and adds two facts** (right-hand panels of Figures 1–3). (i) Training grid points score the same as held-out ones (0.158 vs 0.162 K, scattered split): the model does not fit the training conditions either, the signature of the mean-state collapse. (ii) Between 370 and 1200 m the samples beat the baselines under W_1 in the band and top splits (0.07 vs 0.12 and 0.08 vs 0.17 K at 650 m), because a point prediction is charged for the spread it cannot represent; the model’s total is set by the top 250 m and the bottom two levels, as for the RMSE.
- **Plumbing.** S within 0.003 psu of 35 with no constraint; land exact; ensemble spread $3\times$ the true within-window spread (Figures 4–6).

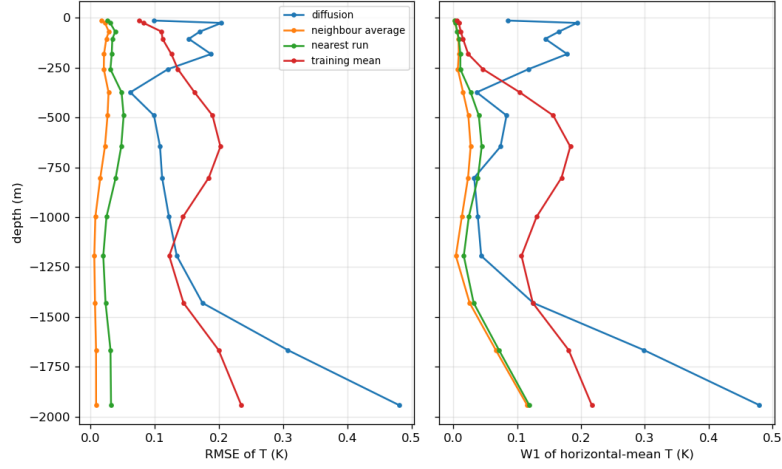
5 Next steps

- Clip: keep the scalar clip’s hold at depth without the corner floor, e.g. per-level bounds from training quantiles (tight where the data are narrow) or the scalar clip loosened only on the levels the corner exceeds.
- Stratification constraint at sampling time (DINO-Fusion’s isotonic projection, on T since S is constant): targets the inversion miss directly.
- Stronger conditioning: classifier-free guidance; year as a third condition (drift of 0.45 K at the bottom inside the window).
- Evaluate against snapshots as well as the time mean (nearest-snapshot RMSE, spread–skill).
- Then: same pipeline on a global Veros configuration (active S , seasonal cycle, multi-century spin-up).

Reproducibility. Branch `miniveros` of the DINO-Fusion repository, package `miniveros/`; commit `1a9470b` for the three runs reported here (training at `2b542aa`, sampling with the per-level clip of `1a9470b`; the scalar-clip rows use `5fdcf01`). Extraction 2.5 min (8 cores), training 17–22 min (one A100), generation of the hold-out and grid sets + evaluation 14–18 min.

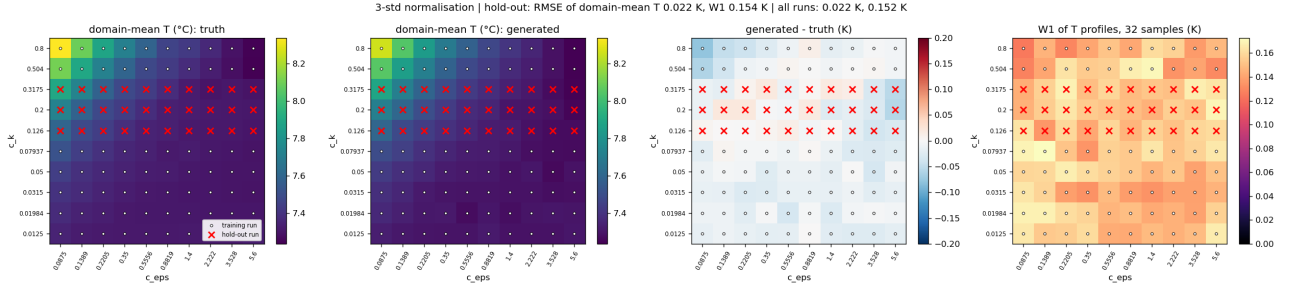


(a) Over the parameter grid: domain-mean T of the truth and of the generated states, their difference, and W_1 of the profile distribution. White dots: training runs; red crosses: held-out runs.

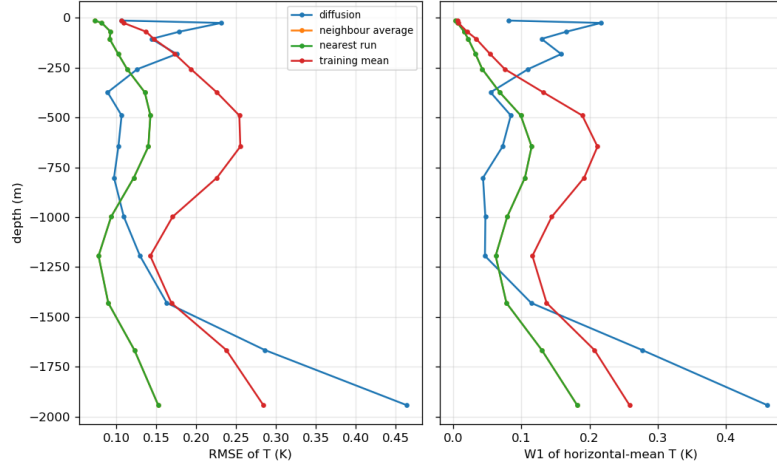


(b) RMSE (left) and W_1 (right) by depth, mean over the held-out runs.

Figure 1: Scattered split (10 interior held-out runs), 32 samples per run.

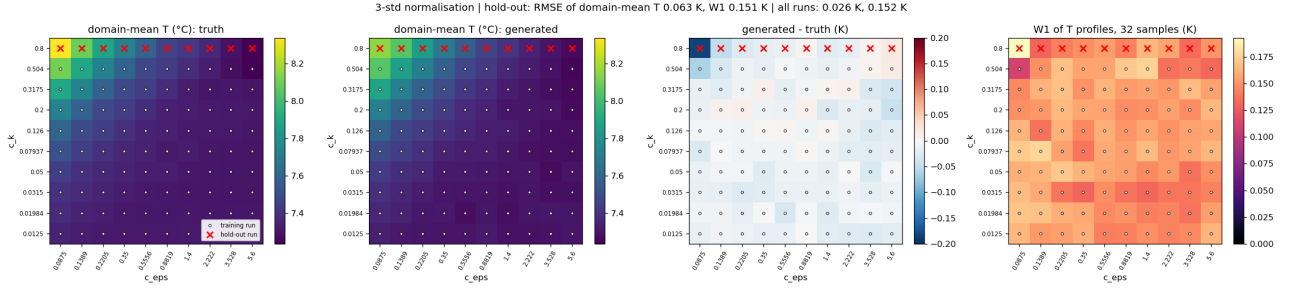


(a) Over the parameter grid: domain-mean T of the truth and of the generated states, their difference, and W_1 of the profile distribution. White dots: training runs; red crosses: held-out runs.

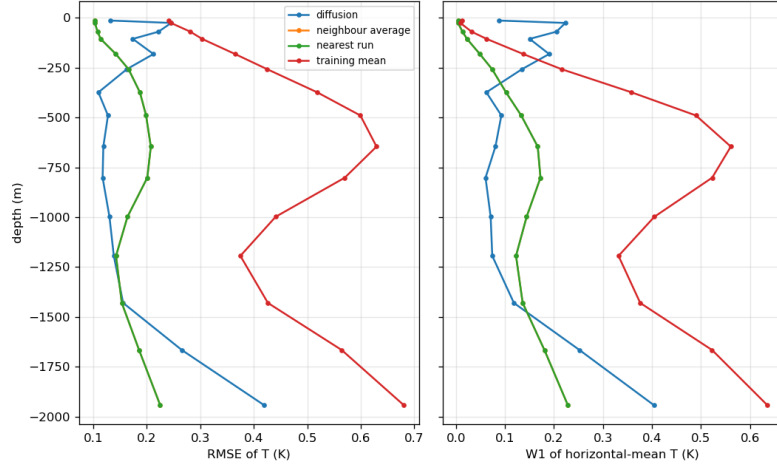


(b) RMSE (left) and W_1 (right) by depth, mean over the held-out runs.

Figure 2: Band of three rows ($c_k = 0.126, 0.2, 0.3175$ held out), 32 samples per run.



(a) Over the parameter grid: domain-mean T of the truth and of the generated states, their difference, and W_1 of the profile distribution. White dots: training runs; red crosses: held-out runs.



(b) RMSE (left) and W_1 (right) by depth, mean over the held-out runs.

Figure 3: Top row ($c_k = 0.8$ held out, extrapolation), 32 samples per run.

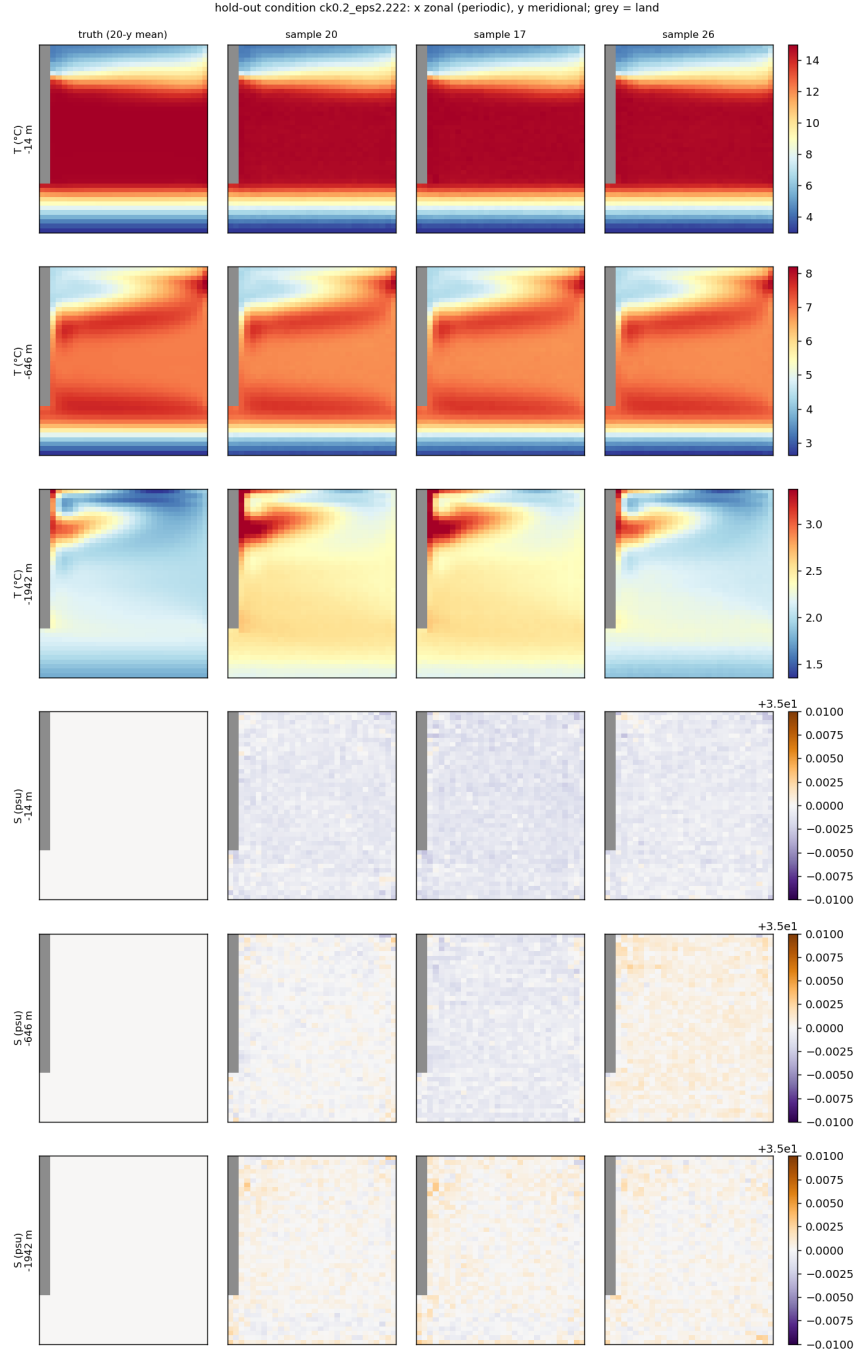


Figure 4: Scattered split, $c_k = 0.2$, $c_\epsilon = 2.222$: true 20-year mean (left) and three random samples at the surface, 650 m and the bottom; T (top three rows), S (bottom three rows). Grey: land.

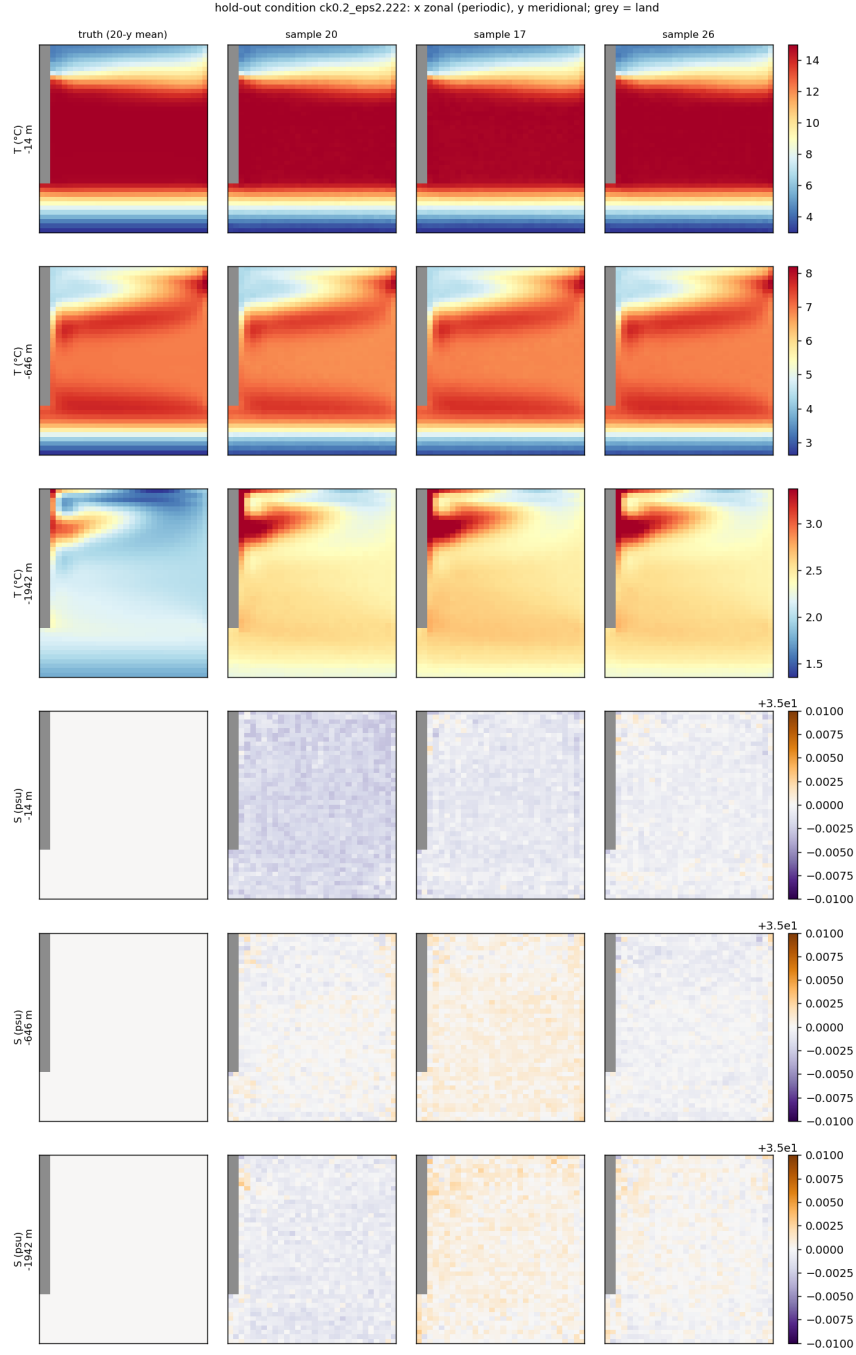


Figure 5: Band-of-three split, same condition ($c_k = 0.2$, $c_\epsilon = 2.222$, now in the middle held-out row).

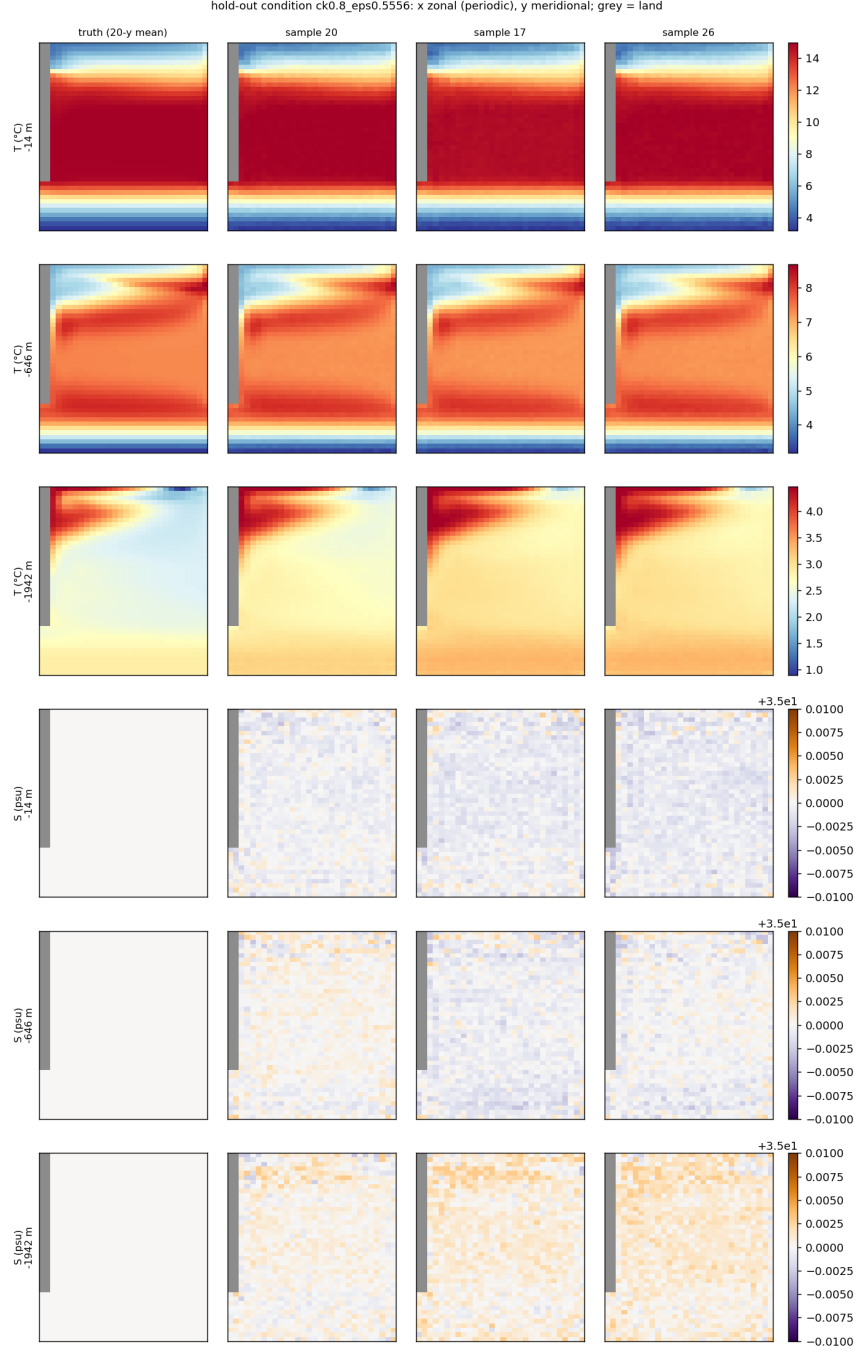


Figure 6: Top-row split, $c_k = 0.8$, $c_\epsilon = 0.5556$ (extrapolation one row above the training grid).